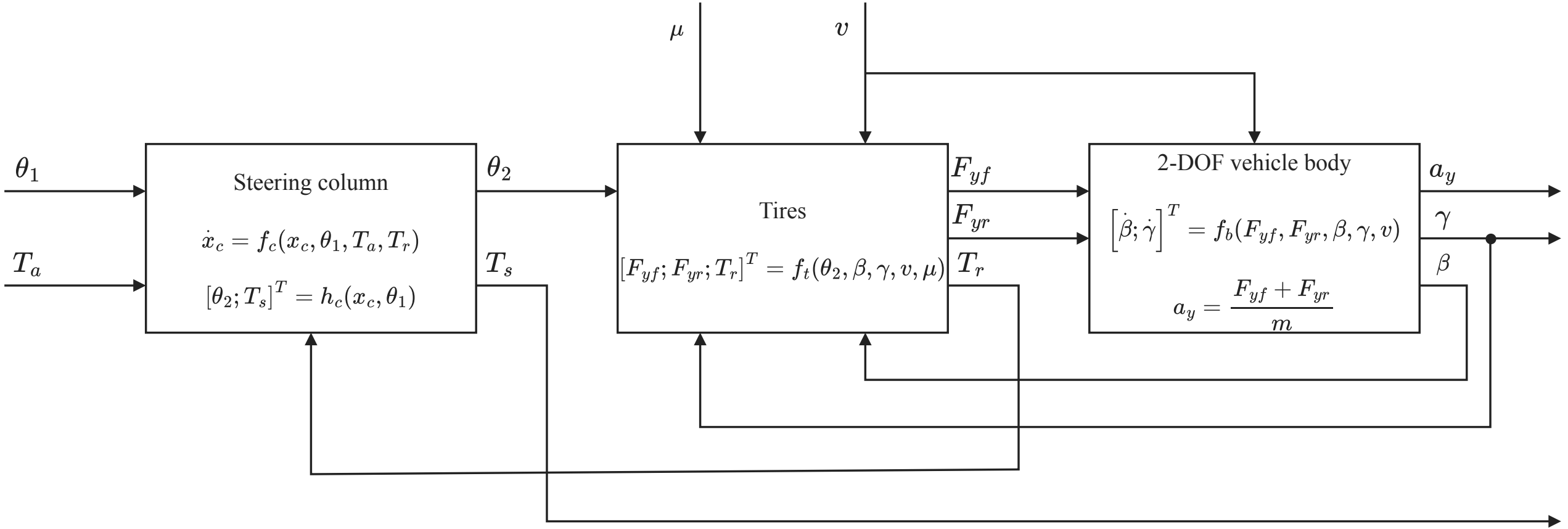
μ
v
θ_1
T_a
Steering column
$\dot{x}_c = f_c(x_c, \theta_1, T_a, T_r)$
$[\theta_2; T_s]^T = h_c(x_c, \theta_1)$
θ_2
T_s
Tires
$[F_{yf}; F_{yr}; T_r]^T = f_t(\theta_2, \beta, \gamma, v, \mu)$
F_{yf}
F_{yr}
T_r
2-DOF vehicle body
$\left[\dot{\beta}; \dot{\gamma}\right]^T = f_b(F_{yf}, F_{yr}, \beta, \gamma, v)$
$a_y = \frac{F_{yf} + F_{yr}}{m}$
a_y
γ
β